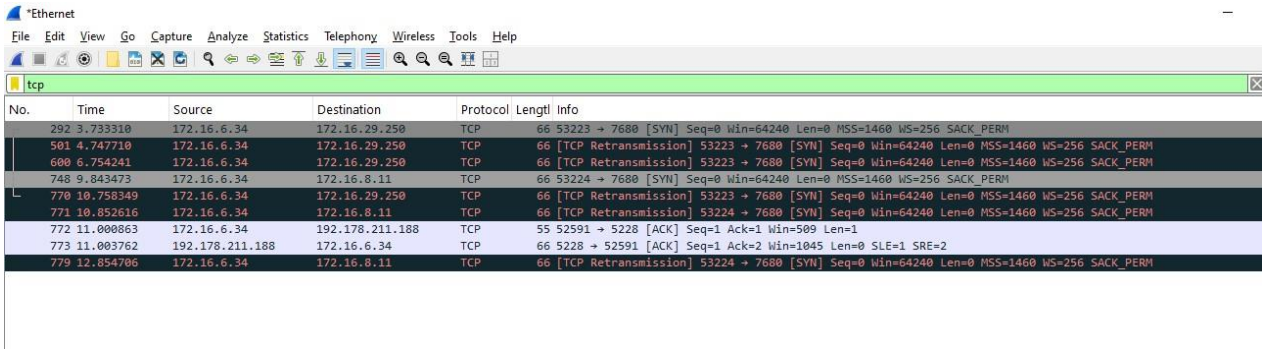
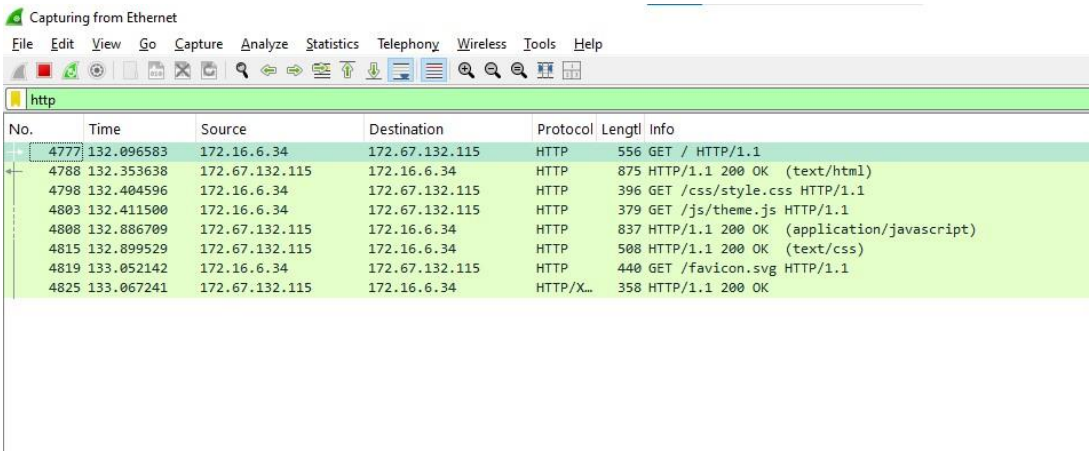
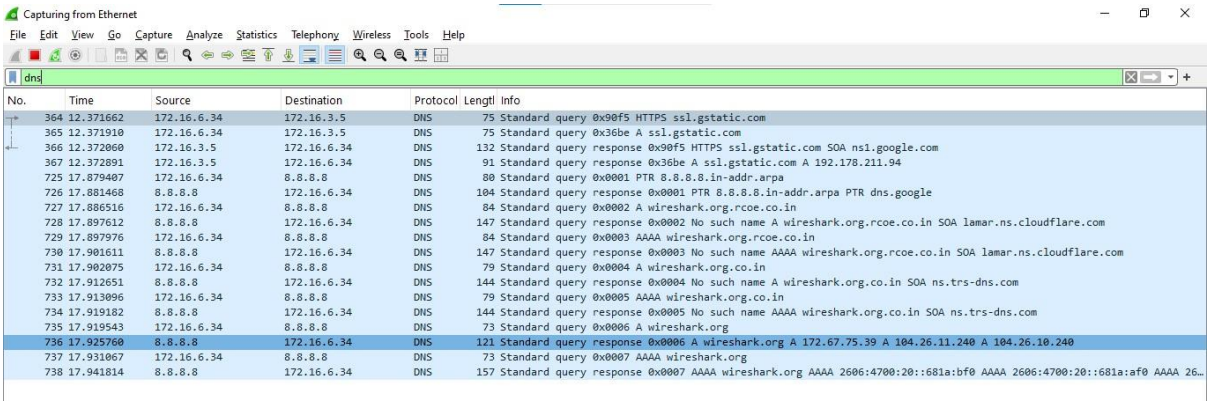
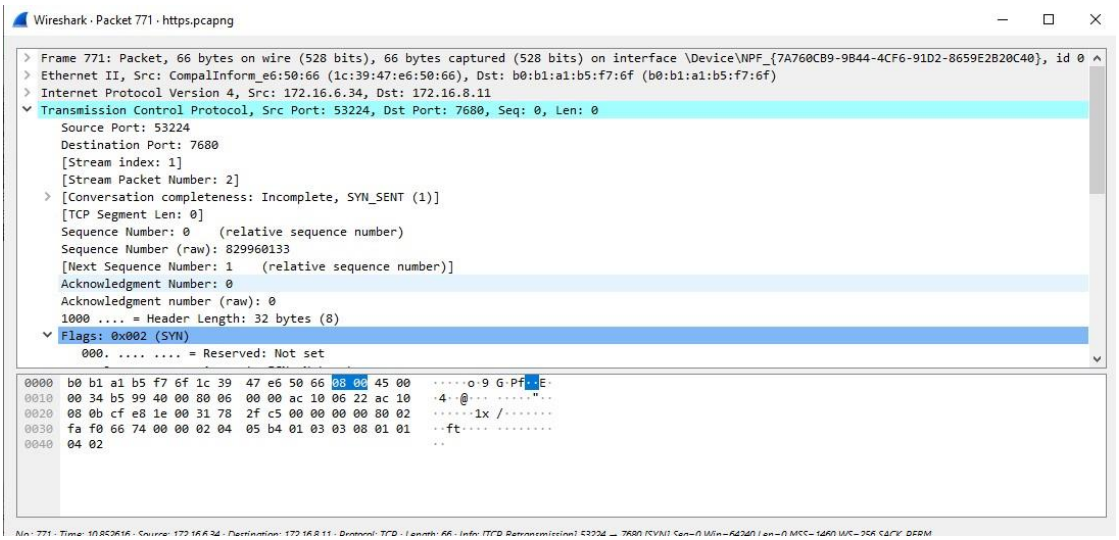
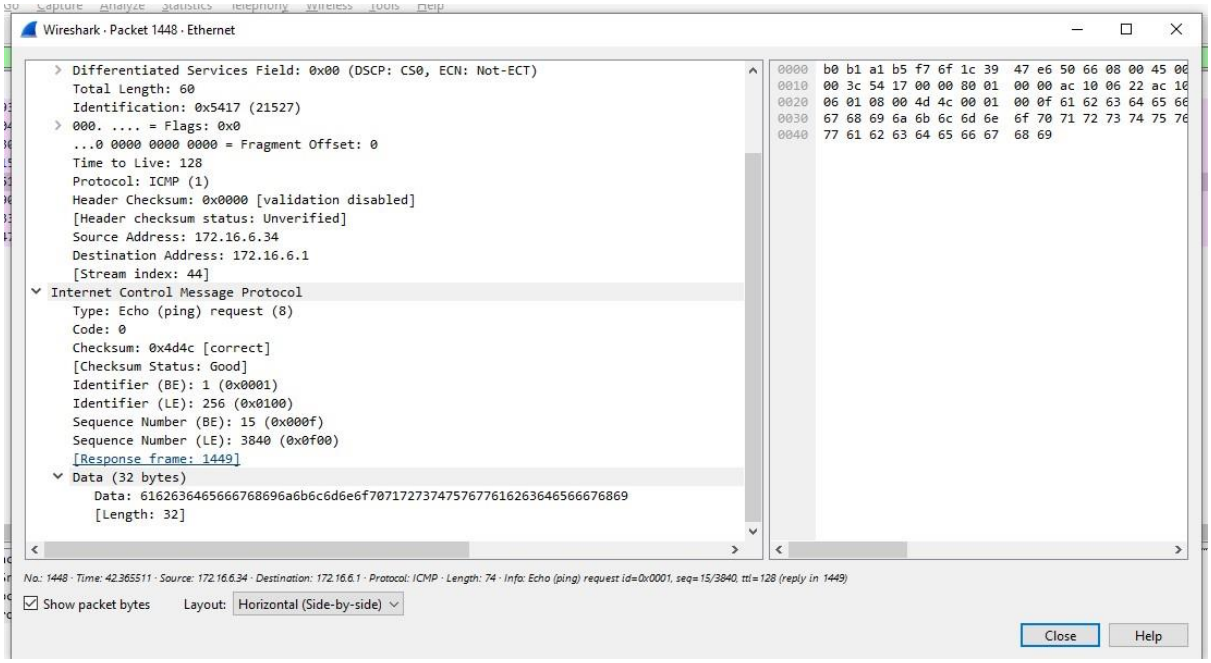
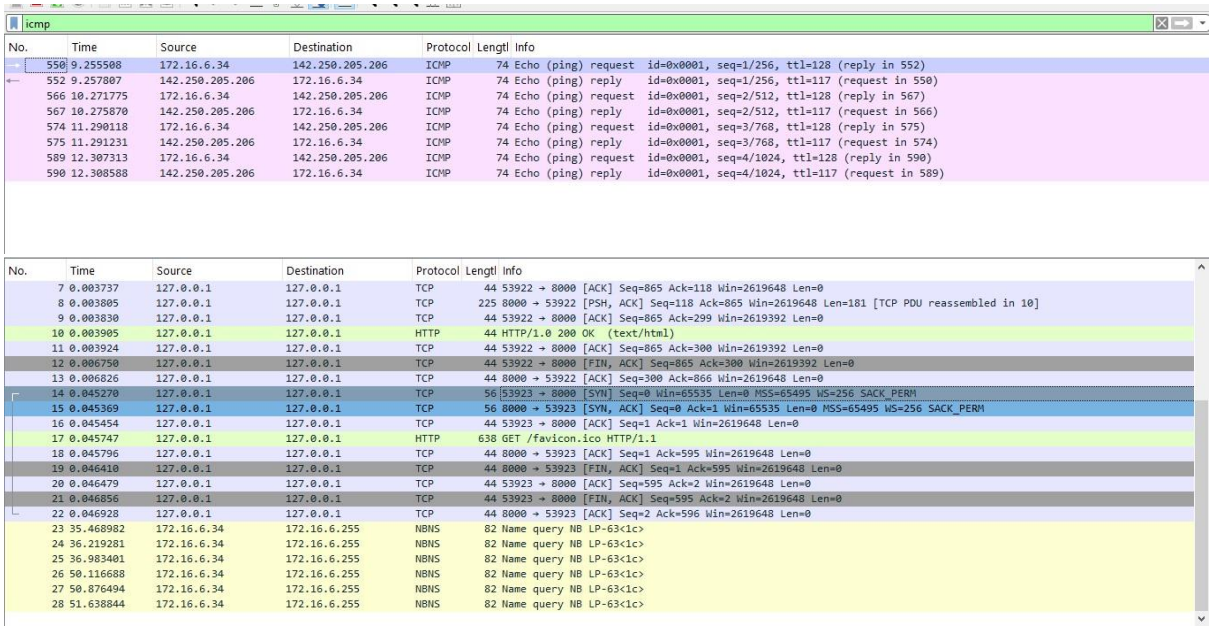

*Ethernet
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
tcp
No. Time Source Destination Protocol Lengtl Info
292 3.733310 172.16.6.34 172.16.29.250 TCP 66 53223 → 7680 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
501 4.747710 172.16.6.34 172.16.29.250 TCP 66 [TCP Retransmission] 53223 → 7680 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
600 6.754241 172.16.6.34 172.16.29.250 TCP 66 [TCP Retransmission] 53223 → 7680 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
748 9.843473 172.16.6.34 172.16.8.11 TCP 66 53224 → 7680 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
770 10.758349 172.16.6.34 172.16.29.250 TCP 66 [TCP Retransmission] 53223 → 7680 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
771 10.852616 172.16.6.34 172.16.8.11 TCP 66 [TCP Retransmission] 53224 → 7680 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
772 11.000863 172.16.6.34 192.178.211.188 TCP 55 52591 → 5228 [ACK] Seq=1 Ack=1 Win=509 Len=1
773 11.003762 192.178.211.188 172.16.6.34 TCP 66 5228 → 52591 [ACK] Seq=1 Ack=2 Win=1045 Len=0 SLE=1 SRE=2
779 12.854706 172.16.6.34 172.16.8.11 TCP 66 [TCP Retransmission] 53224 → 7680 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM

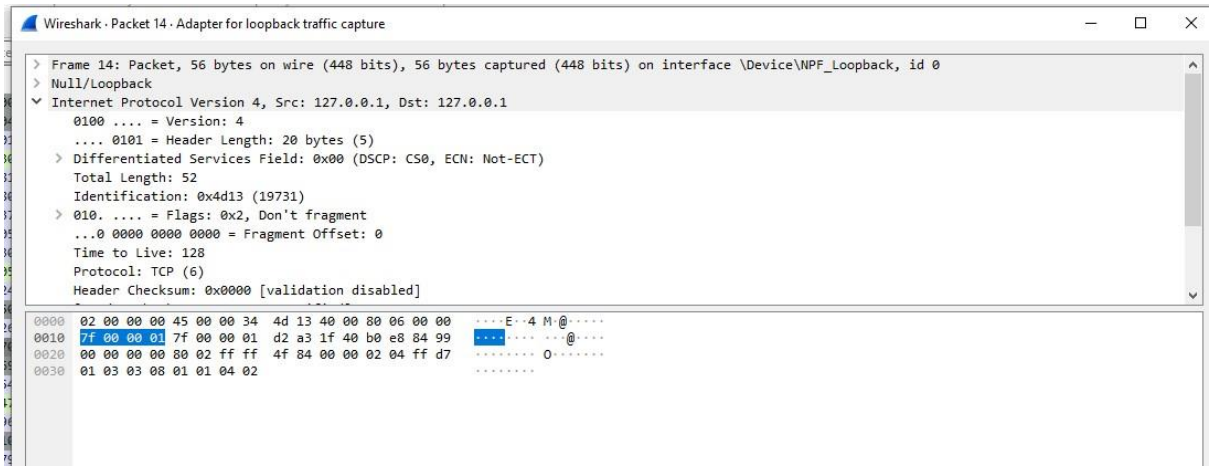

Capturing from Ethernet
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
http
No. Time Source Destination Protocol Lengtl Info
4777 132.096583 172.16.6.34 172.67.132.115 HTTP 556 GET / HTTP/1.1
4788 132.353638 172.67.132.115 172.16.6.34 HTTP 875 HTTP/1.1 200 OK (text/html)
4798 132.404596 172.16.6.34 172.67.132.115 HTTP 396 GET /css/style.css HTTP/1.1
4803 132.411500 172.16.6.34 172.67.132.115 HTTP 379 GET /js/theme.js HTTP/1.1
4808 132.886709 172.67.132.115 172.16.6.34 HTTP 837 HTTP/1.1 200 OK (application/javascript)
4815 132.899529 172.67.132.115 172.16.6.34 HTTP 508 HTTP/1.1 200 OK (text/css)
4819 133.052142 172.16.6.34 172.67.132.115 HTTP 440 GET /favicon.svg HTTP/1.1
4825 133.067241 172.67.132.115 172.16.6.34 HTTP/X… 358 HTTP/1.1 200 OK


Capturing from Ethernet
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
dns
No. Time Source Destination Protocol Lengtl Info
364 12.371662 172.16.6.34 172.16.3.5 DNS 75 Standard query 0x90f5 HTTPS ssl.gstatic.com
365 12.371910 172.16.6.34 172.16.3.5 DNS 75 Standard query 0x36be A ssl.gstatic.com
366 12.372060 172.16.3.5 172.16.6.34 DNS 132 Standard query response 0x90f5 HTTPS ssl.gstatic.com SOA ns1.google.com
367 12.372891 172.16.3.5 172.16.6.34 DNS 91 Standard query response 0x36be A ssl.gstatic.com A 192.178.211.94
725 17.879407 172.16.6.34 8.8.8.8 DNS 80 Standard query 0x0001 PTR 8.8.8.8.in-addr.arpa
726 17.881468 8.8.8.8 172.16.6.34 DNS 104 Standard query response 0x0001 PTR 8.8.8.8.in-addr.arpa PTR dns.google
727 17.886516 172.16.6.34 8.8.8.8 DNS 84 Standard query 0x0002 A wireshark.org.rcoe.co.in
728 17.897612 8.8.8.8 172.16.6.34 DNS 147 Standard query response 0x0002 No such name A wireshark.org.rcoe.co.in SOA lamar.ns.cloudflare.com
729 17.897976 172.16.6.34 8.8.8.8 DNS 84 Standard query 0x0003 AAAA wireshark.org.rcoe.co.in
730 17.901611 8.8.8.8 172.16.6.34 DNS 147 Standard query response 0x0003 No such name AAAA wireshark.org.rcoe.co.in SOA lamar.ns.cloudflare.com
731 17.902075 172.16.6.34 8.8.8.8 DNS 79 Standard query 0x0004 A wireshark.org.co.in
732 17.912651 8.8.8.8 172.16.6.34 DNS 144 Standard query response 0x0004 No such name A wireshark.org.co.in SOA ns.trs-dns.com
733 17.913096 172.16.6.34 8.8.8.8 DNS 79 Standard query 0x0005 AAAA wireshark.org.co.in
734 17.919182 8.8.8.8 172.16.6.34 DNS 144 Standard query response 0x0005 No such name AAAA wireshark.org.co.in SOA ns.trs-dns.com
735 17.919543 172.16.6.34 8.8.8.8 DNS 73 Standard query 0x0006 A wireshark.org
736 17.925760 8.8.8.8 172.16.6.34 DNS 121 Standard query response 0x0006 A wireshark.org A 172.67.75.39 A 104.26.11.240 A 104.26.10.240
737 17.931067 172.16.6.34 8.8.8.8 DNS 73 Standard query 0x0007 AAAA wireshark.org
738 17.941814 8.8.8.8 172.16.6.34 DNS 157 Standard query response 0x0007 AAAA wireshark.org AAAA 2606:4700:20::681a:bf0 AAAA 2606:4700:20::681a:af0 AAAA 26…


Wireshark · Packet 771 · https.pcapng
Frame 771: Packet, 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface \Device\NPF_{7A760CB9-9B44-4CF6-91D2-8659E2B20C40}, id 0
Ethernet II, Src: CompalInform_e6:50:66 (1c:39:47:e6:50:66), Dst: b0:b1:a1:b5:f7:6f (b0:b1:a1:b5:f7:6f)
Internet Protocol Version 4, Src: 172.16.6.34, Dst: 172.16.8.11
Transmission Control Protocol, Src Port: 53224, Dst Port: 7680, Seq: 0, Len: 0
Source Port: 53224
Destination Port: 7680
[Stream index: 1]
[Stream Packet Number: 2]
[Conversation completeness: Incomplete, SYN_SENT (1)]
[TCP Segment Len: 0]
Sequence Number: 0 (relative sequence number)
Sequence Number (raw): 829960133
[Next Sequence Number: 1 (relative sequence number)]
Acknowledgment Number: 0
Acknowledgment number (raw): 0
1000 .... = Header Length: 32 bytes (8)
Flags: 0x002 (SYN)
000. .... = Reserved: Not set
0000 b0 b1 a1 b5 f7 6f 1c 39 47 e6 50 66 08 00 45 00
0010 00 34 b5 99 40 00 80 06 00 00 ac 10 06 22 ac 10
0020 08 0b cf e8 1e 00 31 78 2f c5 00 00 00 00 80 02
0030 fa f0 66 74 00 00 02 04 05 b4 01 03 03 08 01 01
0040 04 02


Wireshark · Packet 1448 · Ethernet
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 60
Identification: 0x5417 (21527)
000. .... = Flags: 0x0
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 128
Protocol: ICMP (1)
Header Checksum: 0x0000 [validation disabled]
[Header checksum status: Unverified]
Source Address: 172.16.6.34
Destination Address: 172.16.6.1
[Stream index: 44]
Internet Control Message Protocol
Type: Echo (ping) request (8)
Code: 0
Checksum: 0x4d4c [correct]
[Checksum Status: Good]
Identifier (BE): 1 (0x0001)
Identifier (LE): 256 (0x0100)
Sequence Number (BE): 15 (0x000f)
Sequence Number (LE): 3840 (0x0f00)
[Response frame: 1449]
Data (32 bytes)
Data: 6162636465666768696a6b6c6d6e6f707172737475767761626364656667686 9
[Length: 32]
0000 b0 b1 a1 b5 f7 6f 1c 39 47 e6 50 66 08 00 45 00
0010 00 3c 54 17 00 00 80 01 00 00 ac 10 06 22 ac 10
0020 06 01 08 00 4d 4c 00 01 00 0f 61 62 63 64 65 66
0030 67 68 69 6a 6b 6c 6d 6e 6f 70 71 72 73 74 75 76
0040 77 61 62 63 64 65 66 67 68 69
No.: 1448 · Time: 42.365511 · Source: 172.16.6.34 · Destination: 172.16.6.1 · Protocol: ICMP · Length: 74 · Info: Echo (ping) request id=0x0001, seq=15/3840, ttl=128 (reply in 1449)
Show packet bytes
Layout: Horizontal (Side-by-side)
Close
Help


icmp
No. Time Source Destination Protocol Lengtl Info
550 9.255508 172.16.6.34 142.250.205.206 ICMP 74 Echo (ping) request id=0x0001, seq=1/256, ttl=128 (reply in 552)
552 9.257807 142.250.205.206 172.16.6.34 ICMP 74 Echo (ping) reply id=0x0001, seq=1/256, ttl=117 (request in 550)
566 10.271775 172.16.6.34 142.250.205.206 ICMP 74 Echo (ping) request id=0x0001, seq=2/512, ttl=128 (reply in 567)
567 10.275870 142.250.205.206 172.16.6.34 ICMP 74 Echo (ping) reply id=0x0001, seq=2/512, ttl=117 (request in 566)
574 11.290118 172.16.6.34 142.250.205.206 ICMP 74 Echo (ping) request id=0x0001, seq=3/768, ttl=128 (reply in 575)
575 11.291231 142.250.205.206 172.16.6.34 ICMP 74 Echo (ping) reply id=0x0001, seq=3/768, ttl=117 (request in 574)
589 12.307313 172.16.6.34 142.250.205.206 ICMP 74 Echo (ping) request id=0x0001, seq=4/1024, ttl=128 (reply in 590)
590 12.308588 142.250.205.206 172.16.6.34 ICMP 74 Echo (ping) reply id=0x0001, seq=4/1024, ttl=117 (request in 589)
No. Time Source Destination Protocol Lengtl Info
7 0.003737 127.0.0.1 127.0.0.1 TCP 44 53922 → 8000 [ACK] Seq=865 Ack=118 Win=2619648 Len=0
8 0.003805 127.0.0.1 127.0.0.1 TCP 225 8000 → 53922 [PSH, ACK] Seq=118 Ack=865 Win=2619648 Len=181 [TCP PDU reassembled in 10]
9 0.003830 127.0.0.1 127.0.0.1 TCP 44 53922 → 8000 [ACK] Seq=865 Ack=299 Win=2619392 Len=0
10 0.003905 127.0.0.1 127.0.0.1 HTTP 44 HTTP/1.0 200 OK (text/html)
11 0.003924 127.0.0.1 127.0.0.1 TCP 44 53922 → 8000 [ACK] Seq=865 Ack=300 Win=2619392 Len=0
12 0.006750 127.0.0.1 127.0.0.1 TCP 44 53922 → 8000 [FIN, ACK] Seq=865 Ack=300 Win=2619392 Len=0
13 0.006826 127.0.0.1 127.0.0.1 TCP 44 8000 → 53922 [ACK] Seq=300 Ack=866 Win=2619648 Len=0
14 0.045270 127.0.0.1 127.0.0.1 TCP 56 53923 → 8000 [SYN] Seq=0 Win=65535 Len=0 MSS=65495 WS=256 SACK_PERM
15 0.045369 127.0.0.1 127.0.0.1 TCP 56 8000 → 53923 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=65495 WS=256 SACK_PERM
16 0.045454 127.0.0.1 127.0.0.1 TCP 44 53923 → 8000 [ACK] Seq=1 Ack=1 Win=2619648 Len=0
17 0.045747 127.0.0.1 127.0.0.1 HTTP 638 GET /favicon.ico HTTP/1.1
18 0.045796 127.0.0.1 127.0.0.1 TCP 44 8000 → 53923 [ACK] Seq=1 Ack=595 Win=2619648 Len=0
19 0.046410 127.0.0.1 127.0.0.1 TCP 44 8000 → 53923 [FIN, ACK] Seq=1 Ack=595 Win=2619648 Len=0
20 0.046479 127.0.0.1 127.0.0.1 TCP 44 53923 → 8000 [ACK] Seq=595 Ack=2 Win=2619648 Len=0
21 0.046856 127.0.0.1 127.0.0.1 TCP 44 53923 → 8000 [FIN, ACK] Seq=595 Ack=2 Win=2619648 Len=0
22 0.046928 127.0.0.1 127.0.0.1 TCP 44 8000 → 53923 [ACK] Seq=2 Ack=596 Win=2619648 Len=0
23 35.468982 172.16.6.34 172.16.6.255 NBNS 82 Name query NB LP-63<1c>
24 36.219281 172.16.6.34 172.16.6.255 NBNS 82 Name query NB LP-63<1c>
25 36.983401 172.16.6.34 172.16.6.255 NBNS 82 Name query NB LP-63<1c>
26 50.116688 172.16.6.34 172.16.6.255 NBNS 82 Name query NB LP-63<1c>
27 50.876494 172.16.6.34 172.16.6.255 NBNS 82 Name query NB LP-63<1c>
28 51.638844 172.16.6.34 172.16.6.255 NBNS 82 Name query NB LP-63<1c>


Wireshark · Packet 14 · Adapter for loopback traffic capture
Frame 14: Packet, 56 bytes on wire (448 bits), 56 bytes captured (448 bits) on interface \Device\NPF_Loopback, id 0
Null/Loopback
Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 52
Identification: 0x4d13 (19731)
010. .... = Flags: 0x2, Don't fragment
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 128
Protocol: TCP (6)
Header Checksum: 0x0000 [validation disabled]
0000 02 00 00 00 45 00 00 34 4d 13 40 00 80 06 00 00
0010 7f 00 00 01 7f 00 00 01 d2 a3 1f 40 b0 e8 84 99
0020 00 00 00 00 80 02 ff ff 4f 84 00 00 02 04 ff d7
0030 01 03 03 08 01 01 04 02

```
C:\wireshark-lab>python lab_server.py
Wireshark laboratory server running...
Open: http://127.0.0.1:8000
127.0.0.1 - - [21/Sep/2026 13:24:59] "GET / HTTP/1.1" 200 -

Received login data:
username=admin&password=admin
Username: admin
Password: admin
127.0.0.1 - - [21/Sep/2026 13:25:27] "POST /login HTTP/1.1" 200 -

Received login data:
username=admin&password=admin
Username: admin
Password: admin
127.0.0.1 - - [21/Sep/2026 13:27:14] "POST /login HTTP/1.1" 200 -
```

Login received

Laboratory test completed.

wireshark-lab > lab_server.py

```python
from http.server import HTTPServer, BaseHTTPRequestHandler
from urllib.parse import parse_qs


class LabServer(BaseHTTPRequestHandler):

    def do_GET(self):
        if self.path == "/":
            html = """
            <html>
            <body>
                <h2>Wireshark Clear-Text Lab</h2>

                <form method="POST" action="/login">
                    Username:
                    <input type="text" name="username"><br><br>

                    Password:
                    <input type="password" name="password"><br><br>

                    <input type="submit" value="Login">
                </form>
            </body>
            </html>
            """

            self.send_response(200)
            self.send_header("Content-Type", "text/html")
            self.end_headers()
            self.wfile.write(html.encode("utf-8"))

    def do_POST(self):
        if self.path == "/login":
            length = int(self.headers.get("Content-Length", 0))
            data = self.rfile.read(length).decode("utf-8")

            print("\nReceived login data:")
            print(data)

            # Parse the form data
```